

EEE 419/591
Summ 2026
Additional Homework 1 – 591 Section

Solve the following additional problem to serve as your extra work for your enrollment in the 591 version of this class. Name your file hw1_591.py and submit it to the designated link on Canvas. Do not add the other problems to this code. If you upload a second attempt to Canvas and it renames your file, ignore the renaming.

In this HW, only limited AI is allowed. Examples of allowed prompts on this HW:

- What is a greatest common divisor of two numbers?
- How do I write an if statement in python? Give me an example
- How do I check if a number is integer in python?
- I got this error message. How do I interpret it?

Examples of prohibited prompts on this HW:

- Write me a code that calculates the greatest common divisor of a list of numbers in python
- I ran this code. [PASTE CODE HERE]. I got this error message. [PASTE THE ERROR MESSAGE HERE]. How do I fix my code?

Do not forget to cite your AI chat if you used any. Check the Syllabus document for details.

Problem 4: Manual DFT Calculations

Calculate the DFT of an array x_{time} . Store it in another array x_{freq} . The DFT is given by

$$x_{\text{freq}}[k] = \sum_{n=0}^{N-1} e^{-\frac{j2\pi kn}{N}} x_{\text{time}}[n] \quad , \quad \forall k = 0, \dots, N-1$$

Write a code that calculates the x_{freq} array using for loops as well as using matrix multiplication. Print max the error, among all N values, to the python built-in FFT package. Use a hard-coded (potentially random) array x_{time} . Do not do Monte Carlo simulations to fine tune the error calculation. Only one error calculation suffices. If the error is zero, increase the precession by changing the format of the numbers used. If the error is still zero, report a zero. Use $N = 512$. A sample print out of your code could look like this

Max error (loop vs FFT): 2.13e-12
Max error (matrix vs FFT): 1.87e-12

MANDATORY – Reminder to cite AI chat